Basic Econometric to Survive Pitt PhD

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Note: This note contains lots of material from textbooks and online materials. I wrote it to prepare Pitt method comprehensive exam, it may contains many typos.

1 Introduction

In the causal inference, we want to answer questions like what would have happened to this person's behavior if she had been subject to an alternative policy T.

Let us define Y_i^T is the average outcomes under treatment and Y_i^C is the average outcomes under control. We are interested in the difference $Y_i^T - Y_i^C$, which is the effect of treatment. The problem is we will never have a subject i both under the treatment and control condition. Therefore, we hope to learn the average effect that is $E(Y_i^T - Y_i^C)$.

Imagine we have data on two group of subjects. One group of people takes the treatment while the other group not. Then we could take the average of both groups and take the difference: $E(Y_i^T|T=1) - E(Y_i^C|T=0)$. We then subtract and add $E(Y_i^C|T=1)$. Finally we have:

$E(Y_i^T|T=1) - E(Y_i^C|T=0) = E(Y_i^T|T=1) - E(Y_i^C|T=1) + E(Y_i^C|T=1) - E(Y_i^C|T=0)$

Where the first two terms in the above equation is the average treatment effect on the treated, this is also the effect we want to know. And the last two terms $E(Y_i^C|T=1) - E(Y_i^C|T=0)$ called selection bias.

This framework is called Potential Outcome Framework.

1.1 Assumptions of Potential Outcome Model

1. Stable Unite Treatment Value Assumption(SUTV).

SUTV as implied by its name is a basic assumption of causal effect stability that requires that the potential outcomes of individuals be unaffected by changes in the treatment exposure of all other individuals. Which means no diffusion effect on others when one get treated.

Typical SUTVA violations share two interrelated features: (1) influence patterns that result from contact across individuals in social or physical space and (2) dilution/ concentration patterns that one can assume would result from changes in the prevalence of the treatment.

When this assumption is violated, we still could say something within a subsample where this assumption is relativly valid.

2. Ignorability (strong version)

Treatment status is independent of the potential outcomes and treatment assignment mechanism.

But more than often, in our observational studies, researchers can not control the treatment assignment.

In the naive estimator of ATE, it has baseline bias and differential treatment effect bias. Only under Independent assignment assumption, the ATE equals to ATT, and ATC.

The independent assumption guarantee that the following assumption are valid: (1). $E(Y_i^T|T=1) = E(Y_i^T|T=0)$. (2).$E(Y_i^C|T=1) = E(Y_i^C|T=0)$, which means there is no selection bias in both the treatment and control group.

If assumption 1 in the above is valid, then ATC is still consistent. If assumption 2 in the above is valid, then ATT is still valid.

The weaker version of ignorability is conditional on variables that affect treatment assignment, the treatment mechanism is independent of potential outcomes.

Even the weaker one is not valid, we could still evaluate whether the subset of the ignorability assumption is valid

How do we eliminate the selection bias so that we can get the average treatment effect? Empirical methods try to solve this problem.

2 Randomized Evaluation

The "Gold Standard" of estimating the ATE is through a randomized experiment. An experiment is a study where assignment to treatment is controlled by the researcher. In the experiment, we randomly divided subjects into treatment and control group where the treatment group take the treatment and controls do not. Then the average treatment effect could be measured by:

$ATE = E(Y_i|T=1) - E(Y_i|T=0)$

Because this is a randomized experiment, the ignorability assumption is valid, there is no selection bias.

There are many ways to do the randomization such as Bernoulli trials (flip coin to decide treatment), completely randomized experiment (sample N_t from population to be treated), Stratified randomized experiment (define blocks according covariates and then random treat in each block), and Pair randomized experiments.

The regression counterpart of the ATE is

$Y_i = \alpha + \delta T + \varepsilon$

then $\hat{\delta}_{\text{OLS, bivariate}} \equiv \frac{\text{Cov}(Y_i, T_i)}{\text{Var}(T_i)} = E\left[Y_i \mid T_i = 1\right] - E\left[Y_i \mid T_i = 0\right]$.

2.1 Problems of Randomized Experiments

The first problem is sometimes doing a experiment have too much financial costs especially experiments that have large scale and involve many labors. The second problem is ethical problems. It is not possible to run all the experiments we would like to because they might affect substantially the economic or social outcomes of the treated population for example, there is no way to assign some countries to be democracy countries while some are not.

More importantly, experiment methods have its own internal and external validity problem. Scholars generally agree that experiments have better internal validity than other methods that based on observational studies. However, experiments could suffer from non response bias, particularly, when people leave the experiment for some reasons systematically related to the outcome, then there is attrition bias. A good example from my own survey study is when people answering political sensitive questions, they tend to not answer or falsify their answers. Although nowadays we have statistical methods and survey method to reduce this concern, it's still a problem. Sometimes, maintaining the allocation to control and treatment to be random is almost impossible. For example, Krueger(2000) evaluation of the Tennessee Star small class size experiment: children were moved to small classes due to parental pressures, bad behavior, etc. The actual class is therefore not random even though the initial assignment was random. It is then important to use the initial assignment as the treatment, because it is the only variation that was randomly assigned. It can then be used as an instrument for actual class size.

In terms of the external validity, experiments are usually taken during a relative short time and limited to a particular geographic area. Therefore, its hard to generalize the results of an experiment to the total population. Even though before the experiment, we could design the experiment subjects to be representative enough to the whole population, small scale experiments do not generate general equilibrium effects that might be very important when policy is applied to everybody in the population.

3 Regression Adjustment as a Strategy to Estimate Causal Effects

Let us first write the potential outcome as the following,

$y^0 = \mu^0 + v^0$ (1)

$y^1 = \mu^1 + v^1$ (2)

then the potential outcome is

$Y = TY^1 + (1-T)Y^0$ (3)

then substitute the first two equation to the third equation, we could have:

$Y = \mu^0 + \left(\mu^1 - \mu^0\right)T + \left\{v^0 + T\left(v^1 - v^0\right)\right\}$ (4)

where v^0 is the baseline bais and $T(v^1 - v^0)$ is the different treatment effect.

In this case, the causal variable T could be correlated with the population level variant of the error term $v^0 + T(v^1 - v^0)$ in Equation (4) in two ways: with v^0 or with $T(v^1 - v^0)$.

Regression analysis as an adjustment technique to estimate a causal effect is to add a sufficient set of "control variables" so that the following assumption meet:

Ignorable Assumption or CIA:

$\left(Y^0, Y^1\right) \perp T \mid X$

or

$\left(v^0, v^1\right) \perp T \mid X$

Therefore,

$Y = \alpha + \delta T + X\beta + \varepsilon^*$ (5)

Regression adjustment by X in equation (5) will yield a consistent and unbiased estimate of the ATE: δ, when the above CIA meet.

In the causal inference, we are interested in the population ATE. In the regression specification in (4), we usually assume the treatment effect is constant, which is the $\mu^1 - \mu^0$ is constant. If however, treatment effect is not constant, then the OLS estimate represents a conditional-variance-weighted estimate of the underlying causal effects of individuals, in which the weights are a function of the conditional variance of T. Under these conditions, the OLS estimate is consistent and unbiased and for this particular weighted average, which is usually not a causal parameter of primary interest.

Second, when we add control variables to regression, the error term will not be $v^0 + T(v^1 - v^0)$ anymore, rather it is a realized error term after controlling for X.

In sum, regression adjustment by X in equation (5) will yield a consistent and unbiased estimate of the ATE when:

1. T is mean independent of and therefore uncorrelated with $v^0 + T(v^1 - v^0)$ conditional on X
2. The causal effect of T does not vary with X
3. A fully flexible parameterization of X is used.

4 More on OLS [1]

Let's first rewrite equation (5) as follow:

$Y = \alpha + X\beta + \varepsilon$ (6)

[1] We assume our data are randomly draw from population for all the following discussion

where Y is the N $\times$ 1 outcome variable and X are N $\times$ K independent variables that include our treatment variable. Then, $\hat{\beta} = (X'X)^{-1} X'Y = \beta + (X'X)^{-1} X'\epsilon$. $\hat{\beta}$ is consistent if $E(\epsilon|X) = 0$ and there is no perfect multicollinearity. Under the assumption of homoscedasticity, $\text{VAR}(\beta) = \sigma_\epsilon^2 (X'X)^{-1}$. When the error term is non-spherical $V(\epsilon) = \Omega$, the asymptotic variance of the OLS estimator is different from the previous formula and is given by: $\text{VAR} = (X'X)^{-1} (X'\Omega X) (X'X)^{-1}$.

The common reasons for non-spherical error term are heteroskedasticity and correlation within groups. In this case we could use robust standard error or cluster standard error to adjust the variance co-variance structure. Failed to do so will provide wrong inference, such as hypotheses test and confidence interval.

4.1 Hypotheses Testing

Under the assumption that error term is normally distributed, $\varepsilon \sim N(0, \sigma_\epsilon^2)$, we could conduct our hypotheses testing.

For a single variable hypotheses testing, we could write the null hypotheses:

$H_0 : \hat{\beta}_k = b$

$\text{SE}(\hat{\beta}_k) = \sqrt{\hat{\Sigma}_{kk}}$

where $\hat{\Sigma}_{kk}$ is the kth element of diagonal of $\hat{\Sigma} = \sigma_\epsilon^2 (X'X)^{-1}$. And, $\hat{\sigma}^2 = \frac{\sum_n (Y - \mathbf{x} \cdot \hat{\beta})^2}{N-K} = \frac{\text{SSE}}{N-K}$. Then the t statistic can be calculated:

$t_{N-K} = \frac{\hat{\beta}_k - b}{\text{SE}(\hat{\beta}_k)} = \frac{\hat{\beta}_k - b}{\sqrt{\hat{\Sigma}_{kk}}}$

So if you have a small sample, the validity of the standard hypothesis test depends on a normality assumption that may or may not be valid, depending on the circumstances.

As our sample size grows large, the sampling distribution of our test statistic is approximately $N(0, 1)$, the normal distribution with mean zero and variance one. This follows from the Central Limit Theorem, and it holds even if ε is not normally distributed.

Based on the t or z(if under large sample size) statistics we get from the above, we could get a p-value, and based on the p-value, we could decide whether we reject or not reject the null hypotheses.

The confidence interval could be constructed based on:

$\text{CI}_{1-\alpha}(\hat{\beta}_k) = [\hat{\beta}_k - z_\alpha \, \text{SE}(\hat{\beta}_k), \hat{\beta}_k + z_\alpha \, \text{SE}(\hat{\beta}_k)]$

If we want to conduct multiple variables hypotheses testing for example: $H_0 : \beta_2 = \beta_3$, then we could write the null hypotheses as:

$H_0 : \mathbf{R}\beta - \mathbf{c} = \mathbf{0}$

where R is a fixed r $\times$ K matrix where r is the number of restriction we intend to test and K is the number of parameters we have in the regression, c is a fixed r $\times$ 1 vector.

We could calculate the Ward statistic:

$W = (\mathbf{R}\beta - \mathbf{c})'(\mathbf{R}\hat{\Sigma}\mathbf{R}')^{-1}(\mathbf{R}\beta - \mathbf{c})$

Again, under small sample size, if we assume a normal error term, the above Ward statistic becomes the F statistic, and is distributed as F(r, N-K).

The above F test is useful to compare the unrestricted model with restricted model, in OLS case, we could test the difference between R^2 is significant or not. Moreover, F test could be used as joint hypothesis testing. For model comparison purpose, we could assume one or multiple of the parameters equal to zero, $H_0 : \hat{\beta}_k = \hat{\beta_{k-1}} = 0$, which we call this restricted model, and then calculate:

$F = \frac{(SSE_R - SSE_U)/J}{SSE_U/(T-K)}$

where SSE is the sum of square error, J is number of restrictions and K is the number of parameters in original model. If we can not reject the null, then this means out unresticted model is not consistent with data. If we reject the null, then our unresticted model is consistent with data compare to restricted ones.

Single Hypotheses Test with Joint Hypotheses Test

When conduct joint hypotheses test or test of the overall significance of the regression model, we should use F test, instead of test each single parameters separately using t test.

$H_0 : \hat{\beta}_k = \hat{\beta_{k-1}} = \hat{\beta_{k-2}} = ... = 0$

$H_1 :$ *One of them or All of them are non zero*

The F test is not just an aggregation of the individual t tests of the coefficients. Two coefficients might each individually be statistically insignificant, yet the Wald test may lead us to reject the null hypothesis that both are zero. Conversely, one of a group of coefficients might be statistically significant, and yet the Wald test may not have us reject the null hypothesis that all are zero.

The F-test that tests both hypotheses simultaneously makes allowance for the fact that the least squares estimators are correlated. It is a test for whether the pair of values $\beta_k = 0$ and $\beta_{k-1} = 0$ are consistent with the data. When separate t-tests are performed, the possibility that $\beta_k = 0$ is not considered when testing $H0 : \beta_{k-1} = 0$, and vice versa. It is not a pair of values being tested with t-tests, but a consequence about a single parameter at a time. Each t test is treated in isolation from the other, no allowance is made for the correlation betweenβ_k and β_{k-1}. As a consequence, the joint F-test at a 5% significance level is not equivalent to separate t-tests that each uses a 5% significance level. Conflicting results can occur. For example, it is possible for individual t-tests to fail to conclude that coefficients are significantly different from zero, while the F-test implies that the coefficients are jointly significant. This situation frequently arises when the data are collinear, where it cause the individual standard error too large that failed to reject null, but F test show parameters are jointly significant.

4.2 Non-Spherical Errors

As we know, the strictly exougeniety of the error term and no perfect multicollinearity guarantee that OLS estimator is consistent and unbiased. In addition, the homoscedasticity assumption garantee that OLS estimator is the minimum-variance unbiased linear estimator. However, more than usualy, the non-spherical error term is not valid due to: 1) heteroskedasticity, $V[\epsilon_i] \neq V[\epsilon_j]$, 2) auto-correlation, $\text{Cov}[\epsilon_i, \epsilon_j] \neq 0$ for some $i \neq j$.

Suppose strict exogeneity holds, but spherical errors fails, with

$$V[\epsilon \,|\, \mathbf{X}] = \sigma^2\Omega = \sigma^2 \begin{bmatrix} \omega_{11} & \omega_{12} & \cdots & \omega_{1N} \\ \omega_{12} & \omega_{22} & \cdots & \omega_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{1N} & \omega_{2N} & \cdots & \omega_{NN} \end{bmatrix}$$

where Ω is known and σ^2 is not known. Then the OLS estimator could be re-write as $\hat{\beta}_{\text{GLS}} = (\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{X}'\Omega^{-1}\mathbf{Y}$, and $V[\hat{\beta}_{\text{GLS}} \,|\, \mathbf{X}] = \sigma^2(\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}$.

Under this condition, the inference such as hypotheses testing and confidence interval based on constant error term will be wrong and we need to use the above estimator from GLS to do the inference.

A special case of GLS is Weighed Least Square where the error term is heteroskedasticity but not correlated with each other:

$$\Omega = \sigma^2 \begin{bmatrix} \omega_{11} & 0 & \cdots & 0 \\ 0 & \omega_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \omega_{NN} \end{bmatrix}$$

The above discussion assumes we know the form of Ω, but if we do not know it, GLS need to change to FGLS.

White's heteroskedasticity-robust estimator:

Let us define,

$$\hat{\mathbf{U}} = \begin{bmatrix} \hat{\varepsilon}_1^2 & 0 & \cdots & 0 \\ 0 & \hat{\varepsilon}_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \hat{\varepsilon}_N^2 \end{bmatrix}$$

Then the "sandwich" form of White's estimator of the OLS variance matrix:

$$\hat{\Sigma}_{\text{HC}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\hat{\mathbf{U}}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1}\left(\sum_{i=1}^{N} \hat{\varepsilon}_i^2 \mathbf{x}_i'\mathbf{x}_i\right)(\mathbf{X}'\mathbf{X})^{-1}$$

White's estimator is consistent but not unbiased, so we may want to apply a sort of bias correction in small samples. A popular choice is the so-called "HC1" estimator, which corrects for the number of parameters estimated the same way the usual OLS variance estimator does:

$\hat{\Sigma}_{\text{HC1}} = \frac{N}{N-K}\hat{\Sigma}_{\text{HC}}$

It is important to know that weighted least squares and heteroskedasticity-consistent standard errors are not mutually exclusive approaches. If you have a suspicion about the error variances that you think can improve the precision of the regression, but you are not totally comfortable with committing to WLS for inference, you can calculate heteroskedasticity-consistent standard errors for a WLS (or GLS) fit.

How to detect heteroskedasticity:

Heteroskedasticity may arise when:

1. The outcome varies across the independent variable
2. Outliers are present in the data
3. You omit one or more independent variables that should be in the model
4. One or more of the independent variable(s) is(are) skewed
5. Incorrect functional form

The first way to detect heteroskedasticity is the plot of residuals against fitted values; or, plot the IV suspected to be correlated with the variance of the error term. In Stata, after running a regression, you could use the rvfplot (residuals versus fitted values) or rvpplot command (residual versus predictor plot)

The second way is to conduct a formal statistical test such as Breusch-Pagan (estat hettest command in Stata after running regression), White's General Test for Heteroskedasticity (command estat imtest, white), Goldfeld-Quandt test, and many other more tests for heteroskedasticity.

Formally, Breusch-Pagan test assume the null hypothesis, $V[\epsilon_i \,|\, \mathbf{X}] = \sigma^2$ for i to N. The test is:

1. Calculate the OLS estimate $\hat{\beta}_{\text{OLS}}$
2. Calculate the OLS residuals, Let $\hat{\varepsilon}$ be the vector of squared residuals
3. Run a regression of $\hat{\varepsilon}$ on Z an N $\times$ Q matrix of covariates. Let $R^2_{\hat{\varepsilon}}$ be the R^2 of this regression
4. Reject the null hypothesis if N$R^2_{\hat{\varepsilon}}$ exceeds the critical value for a χ^2_{q-1} distribution.

In the canonical version of this test, Z is equal to X, while White's General Test is more powerful in which Z contains each variable in X as well as all second-order terms (squares and interactions).

How to correct heteroskedasticity

1. Respecify the Model/Transform the Variables if heteroskedasticity results from improper model specification

2. Use heteroskedasticity-robust estimator we just mentioned above

3. Use Weighted Least Squares

How to detect auto-correlation

Serial correlation usually happened when observations draw from clusters, observations from the same clusters are usually correlated. Also, in the panel data and time series data, auto-correlation is unavoidable.

Serial correlation will not affect the unbiasedness or consistency of OLS estimators, but it does affect their efficiency. With positive serial correlation, the OLS estimates of the standard errors will be smaller than the true standard errors. This will lead to the conclusion that the parameter estimates are more precise than they really are.

If we assume errors are homoskedastic, and every group have n observations. In the presence of auto-correlation within clusters, the $V[\hat{\beta} \,|\, \mathbf{X}] = (1+(n-1)\rho)\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$, where ρ is the correlation between error terms. Moulton factor is $\sqrt{1+(n-1)\rho}$. The last part is the variance from OLS. This means the OLS standard error calculation will underestimate the true standard errors by a factor of $\sqrt{1+(n-1)\rho}$. If for some reason we were certain the homoskedasticity assumption held and we knew ρ, then we could simply correct the OLS standard errors by applying this formula. More broadly, if we knew the exact structure of heteroskedasticity and autocorrelation, we could obtain an efficient estimate of β and proper standard errors by using GLS. Usually, however, this option will not be available to us.

To detect auto-correlation, we could use auto-correlation plot (ac or pac command in Stata). We could also conduct Durbin –Watson test under the null hypotheses that there is no auto-correlation,

$d = \frac{\sum_{t=1}^{n-1}(\hat{\varepsilon}_{t+1}-\hat{\varepsilon}_t)^2}{\sum_{t=1}^{n}\hat{\varepsilon}_t^2}$

Upper and lower critical values, d_U and d_L have been tabulated for different values of K (the number of explanatory variables) and N. If $d < d_L$ then reject null, If $d > d_U$ then do not reject, if $d_L < d < d_U$, test is inconclusive.

For Durbin's alternative test and the Breusch–Godfrey test that null hypothesis that there is no pth order serial correlation

1. Calculate the OLS estimate $\hat{\beta}_{\text{OLS}}$

2. Calculate the OLS residuals, Let $\hat{\varepsilon}_t$ be the vector of squared residuals

3. Regression of the residuals $\hat{\varepsilon}_t$ against p lags of $\hat{\varepsilon}_t$ and on all the covariates in the original regression. And calculate NR^2 from this regression.

4. Durbin's alternative test is computed by performing a Wald test to determine whether the coefficients of pth lags are jointly different from zero, assume a χ_p^2 distribution.

5. The Breusch–Godfrey test is computed as $\mathrm{N}R^2$ as we get from above. Breusch–Godfrey test is asymptotically distributed χ^2_p.

Clustered Standard Errors

We assume heteroskedasticity of unknown form, within and across groups, Autocorrelation of unknown form within groups, no autocorrelation across groups.

The cluster standard error could be write as,

$$\Omega = \begin{bmatrix} \Omega_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \Omega_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \Omega_G \end{bmatrix}$$

where Ω_g is a symmetric N $\times$ N matrix. These may differ from each other—each group may have its own special auto-correlation structure.

If we knew Ω_g exactly, we could use GLS to efficiently estimate β and obtain correct standard errors. That is unlikely. Instead, we will use OLS to obtain our coefficient estimates, then we will correct the standard errors so as to be approximately correct in large samples. The cluster-robust variance estimator is

$\hat{\Sigma}_{\mathrm{CR}} = (\mathbf{X}^\top \mathbf{X})^{-1} \left(\sum_{g=1}^{G} \mathbf{X}_g^\top \hat{e}_g \hat{e}_g^\top \mathbf{X}_g \right) (\mathbf{X}^\top \mathbf{X})^{-1}$

where $\hat{e}_g$ is the N $\times$ 1 vector of OLS residuals for the gth groups.

Also like White's estimator for heteroskedasticity, the cluster-robust estimator is consistent, but not unbiased. It approaches the truth as the number of groups grows large (holding fixed n, the number of observations per group), but it may be badly biased in small samples. Therefore, in the small sample, when we use clustered standard error we need to adjust the bias.

When to use cluster standard error

Cameron and Miller (2015) paper suggest that when there is within-cluster correlation of the regressors, within-cluster correlation of the error, and when there are number of observations in each cluster. "There are settings where one may not need to use cluster-robust standard errors. We outline several though note that in all these cases it is always possible to still obtain cluster-robust standard errors and contrast them to default standard errors. If there is an appreciable difference, then use cluster robust standard errors". Abadie et al (2017) paper gives different suggestions. They argues that even there is no correlation within the group, cluster standard errors still matter a lot. They suggest We should cluster: in the presence of heterogenous treatment effect and small number of clusters compared to the overall population. Also, when there is correlation between treatment and clusters (cluster assignment). We should not cluster: In pure randomized control trial (or any situation without sampling clustering or assignment clustering) and when there is constant treatment effect and no clustering in the assignment.

4.3 Residual Non-Normality

In the small sample analysis, we assume error terms are normally distributed, but in the large sample, error term are asymptotically normally distributed by central limited theory.

Residual non-normality arise mostly because:

1. Distributions of the dependent and/or independent variables are themselves significantly non-normal
2. The linearity assumption is violated
3. There are two or more subsets of the data having different statistical properties
4. Outliers

How to detect non-normality

The easiest way to detect residual non-normality is residual Plots after running the regression (kdensity in Stata) or normal quantile plot of the residuals (qnorm in Stata). We could also conduct Shapiro–Wilk normality test (swilk in Stata), to detect outliers we could use avplots in Stata.

How to correct non-normality

If the non-normality caused by the first two problem, nonlinear transformation of variables will most of the time solve the problem. If it is caused by the third reason, then separate models should be built, or else some data should merely be excluded, provided that there is some a prior criterion that can be applied to make this determination. If it is caused by the forth reason, then we need to detect which data points are outliers and closely examine why they are outliers. Are they genuine (i.e., not the result of data entry errors), are they explainable, are similar events likely to occur again in the future, and how influential are they in your model-fitting results? If they are merely errors or if they can be explained as unique events not likely to be repeated, then you may have cause to remove them. In some cases, however, it may be that the extreme values in the data provide the most useful information about values of some of the coefficients and/or provide the most realistic guide to the magnitudes of forecast errors.

4.4 Functional Form Misspecification

In the common regression model, we assume the dependent variable and independent variables have linear relationship in the parameters and additivity. But we may violate these assumptions. If you fit a linear model to data which are nonlinearly or nonadditively related, your predictions are likely to be seriously in error, especially when you extrapolate beyond the range of the sample data.

Models are non-linear or non-additivity when

1. Dependent variable is not continuous, it is categorical variables

2. There is interaction relation in independent variables

3. Non-monotonic relationship between dependent and independent variables, such as quadratic terms

4. Some covariate need to take the natural logarithm

How to detect misspecification

First of all, we need to examine whether our dependent variable is continuous or categorical variables. Second, non-linearity is usually most evident in a plot of observed versus predicted values or a plot of residuals versus predicted values. The points should be symmetrically distributed around a diagonal line in the former plot or around horizontal line in the latter plot, with a roughly constant variance. (The residual-versus-predicted-plot is better than the observed-versus-predicted plot for this purpose, because it eliminates the visual distraction of a sloping pattern.) Look carefully for evidence of a "bowed" pattern, indicating that the model makes systematic errors whenever it is making unusually large or small predictions. In multiple regression models, nonlinearity or nonadditivity may also be revealed by systematic patterns in plots of the residuals versus individual independent variables.

Beyond statistical method, we also need to think deeply about the relationship between our dependent variable and independent variables theoretically. More than usual, our theories and hypotheses imply that variables in our studies should have non-linear relationship. A classical example is the relationship between inequality and democratization, where they have a inverted U relationship.

How to correct functional form misspecification

Consider applying a nonlinear transformation to the dependent and/or independent variables if you can think of a transformation that seems appropriate. For example, if the data are strictly positive, the log transformation is an option. If a log transformation is applied to the dependent variable only, this is equivalent to assuming that it grows (or decays) exponentially as a function of the independent variables. If a log transformation is applied to both the dependent variable and the independent variables, this is equivalent to assuming that the effects of the independent variables are multiplicative rather than additive in their original units. This means that, on the margin, a small percentage change in one of the independent variables induces a proportional percentage change in the expected value of the dependent variable, other things being equal. Models of this kind are commonly used in modeling price-demand relationships, as illustrated on the beer sales example on this web site.

Another possibility to consider is adding another regressor that is a nonlinear function of one of the other variables. For example, if you have regressed Y on X, and the graph of residuals versus predicted values suggests a parabolic curve, then it may make sense to regress Y on both X and X^2. The latter transformation is possible even when X and/or Y have negative values, whereas logging is not. Higher-order

terms of this kind (cubic, etc.) might also be considered in some cases. But don't get carried away! This sort of "polynomial curve fitting" can be a nice way to draw a smooth curve through a wavy pattern of points (in fact, it is a trend-line option on scatterplots on Excel), but it is usually a terrible way to extrapolate outside the range of the sample data.

Finally, it may be that you have overlooked some entirely different independent variable that explains or corrects for the nonlinear pattern or interactions among variables that you are seeing in your residual plots. In this cases, you need to draw sights from existing literature that related to the questions you study to see if there any variables or relationship between vairables that are important for modeling but missing in your research.

4.5 Omitted Variable Bias

Omitted variable bias is the bias in the OLS estimator that arises when the regressor, X, is correlated with an omitted variable. For omitted variable bias to occur, two conditions must be fulfilled: 1) X is correlated with the omitted variable. The omitted variable is a determinant of the dependent variable Y. A classical case is the return of education where ability is the omitted variable.

Note, if a omitted variable have effect on dependent variable but does not correlated with included variables or it correlated with included variables but have no effect on dependent varaiable, then there is no OVB

Suppose our true model is

$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$

but we estimate

$Y = \alpha^* + \beta_1^* X_1 + \varepsilon^*$

then

$\beta_1^* = \frac{\hat{C}ov(X_1,Y)}{\hat{V}(X_1)}$
$= \frac{\hat{\text{Cov}}(X_1,a+\beta_1 X_1+\beta_2 X_2+e)}{\hat{V}(X_1)}$
$= \frac{\hat{\text{Cov}}(X_1,a)+\beta_1\hat{\text{Cov}}(X_1,X_1)+\beta_2\hat{\text{Cov}}(X_1,X_2)+\hat{\text{Cov}}(X_1,e)}{\hat{V}(X_1)}$
$= \frac{0+\beta_1\hat{V}(X_1)+\beta_2\hat{C}ov(X_1,X_2)+0}{\hat{V}(X_1)}$
$\beta_1^* = \beta_1 + \beta_2\frac{\hat{C}ov(X_1,X_2)}{\hat{V}(X_1)}$

In general Short regression coefficient = Long regression coefficient + [coefficient on omitted variables in long regression] + [coefficient on omitted variables when regressed on included variables]

This formula is very useful in determining the sign of the omitted variables bias. For instance, in the returns to education example with ability as the omitted variable, we expect that unobserved ability will have a positive impact on wages in the long regression. If we assume that higher ability people choose to

get more schooling, then the omitted variables bias is positive, which means that our estimated coefficient on schooling is biased upwards.

From the above formula, we know β_1^* is a biased estimator of β_1. Further, this bias will not disappear as sample size gets larger, so the omission of a variable from a model also leads to an inconsistent estimator.

How to detect OVB

One way to find whether we have OVB is to ask ourselves: do I include all the theoretical relevant variables based on the existing literature that related to my research questions? Often times, theories could give us some insights of potential missing variables that we need to include.

In Stata, we could use ovtest after running regression. The null hypothesis is that the model does not have omitted-variables bias. Another way is to check the balance of of our covariates under different treatment status. If they are different significantly, then it is very likely that there is a confounder variable.

We now have many popular way to estimate the extent of omitted variables bias(Imbens 2003; Altonji, Elder and Taber 2005; Cinelli and Hazlett 2020;), which we call sensitivity analysis. These methods intend to answer questions such as how strong would a particular confounder (or group of confounders) have to be to change our conclusions?

The idea is, suppose we have a binary treatment variable D=0,1, then we can compute the normalized shift due to observables:

$$\frac{E\big(X'\beta|D{=}1\big)-E\big(X'\beta|D{=}0\big)}{\mathrm{Var}(X'\beta)}$$

and ask how large the normalized shift due to unobservables

$$\frac{E(\epsilon|D{=}1)-E(\epsilon|D{=}0)}{\mathrm{Var}(\epsilon)}$$

would have to be in order to explain away the entire estimate of β. If selection on unobservables has to be very large compared to selection on observables in order to attribute all our results to omitted variables bias, we feel more confident about our results.

How to correct OVB

One way to correct OVB if OVB is caused by missing some observable variables, then we could direct add those variable to equations. If, however, OVB is caused by variables that is hard to measure or unobservale at all, then we could find proxy variables or using other research design such as instrument variable approach.

Over-controlling and Post treatment Bias

Contrary to under controlling where we will have OVB, overall controlling or specifically when we control variables that lies along your causal path way, after the treatment variable but before your outcome variable, we will have post treatment bias. Formally, if we have

$$\begin{tikzcd} \text{Treatment} \arrow[r] \arrow[d] & \text{Outcome} \\ \text{Mediator} \arrow[ur] & \end{tikzcd}$$

where a mediator is a post treatment variable that changes the effect of treatment. For example, if we believe democratization increase GDP and then decrease civil conflict, then in this case, democratization is the treatment variable, GDP is the mediator and civil conflict is the outcome. Treatment effect should equal to the direct effect of democratization plus the effect through mediator. When we study the effect of democratization on civil conflict and we control GDP, we will have post treatment bias. Because conditioning on a post treatment variable changes the quantity of interest from an overall average treatment effect to a direct effect of the treatment net the post treatment variable. A more severe consequence of conditioning on post treatment variables, which is selection bias that leaves the coefficient of interest without any causal interpretation. Selection bias of this kind occurs because the post treatment variable could be endogenous and related to the outcome in noncausal ways. When this occurs, conditioning on this variable can induce spurious correlations between the treatment and the outcome.

Post treatment bias usually happens when we are trying to control a seemingly confounding variable but it is actually a post treatment variable or when we want to do causal mediation analysis. For the second reason, we need to rely on other method such as causal mediation analysis developed by Imai, Keele, and Yamamoto(2010).

Another form of over controlling is you add some irrevalant variables. In this case, β is still consistent, but the variance is inflated.

4.6 The measurement error problem

Assume we want to estimate:

$y_i^* = \alpha + \beta x_i^* + \varepsilon_i$

But for some reason, we can not perfect observe y^*, x^*. What we could observe is y, x, and

$y_i = y_i^* + v_i$

$x_i = x_i^* + w_i$

In the "classical" measurement error case, the assumption is that measurement errors are uncorrelated with the truth, and with each other:

$E\left[v_i y_i^*\right] = E\left[w_i x_i^*\right] = E\left[v_i x_i^*\right] = E\left[w_i y_i^*\right] = E\left[v_i w_i\right] = 0$

and we also assume $E\left[\epsilon_i x_i^*\right] = 0$

To re-write the model in variables we actually observe:

$y_i - v_i = \alpha + \beta(x_i - w_i) + \varepsilon_i$

$y_i = \alpha + \beta x_i + (\varepsilon_i - \beta w_i + v_i)$

It is easy to see that the new error term in the parenthesis are correlated with x which violated the strict exougenous assumption. If we estimate the above wrong equation, we have

$\hat{\beta} = \frac{cov(x_i, y_i)}{var(x_i)}$ and it is easy to show that

$\text{Plim}\hat{\beta} = \beta \frac{\sigma^2_{x^*}}{\sigma^2_{x^*} + \sigma^2_w}$

Therefore, when we have measurement error, there is an attenuation bias: the estimated coefficient is smaller than the true coefficient. The measurement error in y does not lead to attenuation bias, in the uncorrelated case. The larger the variance of the error term relative to the variance of the underlying variable (the 'signal to noise ratio'), the larger the attenuation bias.

The problem of measurement error is even server when we use the first difference estimation. Generally,

$\text{Plim}\, \hat{\beta}_{FD} = \beta \frac{\sigma^2_{\Delta x^*}}{\sigma^2_{\Delta x^*} + \sigma^2_{\Delta w}}$

If measurement errors are independent measurement error in each period (an extreme case), then $\sigma^2_{\Delta w} = 2\sigma^2_w$. However, x^* is presumably strongly autocorrelated, so $\sigma^2_{\Delta X^*} < 2\sigma^2_{X^*}$. Therefore the attenuation bias is stronger. In fact, it can become really large if the underlying variable does not move very much over time but there is measurement error in every period.

It is also possible that measurement error does not satisfy the "classical"" measurement error formula. It is possible that the error are correlated with x, in this case, measurement error need not lead to attenuation bias. It depends on how large is the correlation between x and error. Sometimes, we even cannot determine how bias is our estimation if the measurement error is non-classical.

How to deal with it

Structure equation modeling has the advantage under the presence of measurement error. We could also use instrument variable approach.

4.7 Simultaneity or Reverse Causality

Simultaneity arises when at least one of the explanatory variable is determined simultaneously alone with y. If say X_k is determined partly as a function of y, then X_k and ϵ are generally correlated. For example, if y is a city murder rate and X_k is size of the police force, size of the police force is partly determined by the murder rate. Conceptually, this is a more difficult situation to analyze because we must be able to think of a situation where we could vary X_k exougenously even though in the data that we collect y and X_k are generally simultaneously.

Experimental methods are useful in this case since in the experiment there is clear intervention and we could measure the outcome before and after the intervention. Instrumental variable is also a way to solve this problem.

4.8 Collinearity

When data are the result of an uncontrolled experiment many of the economic variables may move together in systematic ways. Such variables are said to be collinear, and the problem is labeled collinearity, or multicollinearity when several variables are involved. In this case there is no guarantee that the data will be "rich in information," nor that it will be possible to isolate the economic relationship or parameters of interest.

The Statistical Consequences of Collinearity

Whenever there are one or more exact linear relationships among the explanatory variables, then the condition of exact collinearity, or exact multicollinearity, exists. In this case the least squares estimator is not defined. Full rand of XX in OLS estimator is violated.

When nearly exact linear dependencies among the explanatory variables exist, some of the variances, standard errors and covariances of the least squares estimators may be large. Large standard errors for the least squares estimators imply high sampling variability, estimated coefficients that are unstable to small changes in the sample or model specification, interval estimates that are wide, and relatively imprecise information provided by the sample data about the unknown parameters.

When estimator standard errors are large, it is likely that the usual t-tests will lead to the conclusion that parameter estimates are not significantly different from zero. This outcome occurs despite possibly high R^2 or "F-values" indicating "significant" explanatory power of the model as a whole. The problem is that collinear variables do not provide enough information to estimate their separate effects, even though economic theory may indicate their importance in the relationship.

Estimates may be very sensitive to the addition or deletion of a few observations, or the deletion of an apparently insignificant variable.

Despite the difficulties in isolating the effects of individual variables from such a sample, accurate forecasts may still be possible if the nature of the collinear relationship remains the same within the new (future) sample observations. For example, in an aggregate production function where the inputs labor and capital are nearly collinear, accurate forecasts of output may be possible for a particular ratio of inputs but not for various mixes of inputs.

How to detect collinearity

The analysis exhibits the signs of multicollinearity — such as, estimates of the coefficients vary excessively from model to model. The t-tests for each of the individual slopes are non-significant (P ¿ 0.05), but the overall F-test for testing all of the slopes are simultaneously 0 is significant (P ¡ 0.05). The correlations among pairs of predictor variables are large.

Looking at correlations only among pairs of predictors, however, is limiting. It is possible that the

pairwise correlations are small, and yet a linear dependence exists among three or even more variables. In this case, we could use variance inflation method,

In particular, the variance inflation factor for the jth predictor is:

$VIF_j = \frac{1}{1-R_j^2}$

where R_j^2 is the R^2-value obtained by regressing the jth predictor on the remaining predictors. A VIF of 1 means that there is no correlation among the jth predictor and the remaining predictor variables, and hence the variance of β_j is not inflated at all. The general rule of thumb is that VIFs exceeding 4 warrant further investigation, while VIFs exceeding 10 are signs of serious multicollinearity requiring correction.

5 Matching

Another way of identifying a causal effect is by matching. The critical assumption of matching method is treatment effect is random conditional on a set of observable variables X, which we could formally define as the following ignorability assumption:

Recall that $ATE = E(Y_i^T|T=1) - E(Y_i^C|T=1) + E(Y_i^C|T=1) - E(Y_i^C|T=0)$. Under the assumption of conditional independent of treatment assignment, we have

$E\left[Y^T \mid T=1, X\right] = E\left[Y^T \mid T=0, X\right]$

$E\left[Y^C \mid T=1, X\right] = E\left[Y^C \mid T=0, X\right]$

In general, if a stratifying variables X completely account for all systematic difference between those who take the treatment and who do not, then conditional on X estimators yield consistent and unbiased estimators of the ATE conditional on a particular value of x of X:

$\{E_N\left[y_i \mid T_i=1, s_i=s\right] - E_N\left[y_i \mid T_i=0, s_i=s\right]\} \quad \xrightarrow{p} E\left[Y^1 - Y^0 \mid X=x\right] = E[\delta \mid X=x]$

and finally the ATE is,

$E_x\left\{E[\delta \mid X=x]\right\} = \int E[\delta \mid X=x] P(X=x|T)$

This means that, if X takes discrete values, we can compare Treatment and Control in all the cells formed by the combination of the Xs, and then take a weighted average over these cells, using as weights the proportion of treated in the cells You could also substitute $P(X=x|T=1)$ or $P(X=x|T=0)$ to calculate the consistent estimator of ATT or ATC.

One thing need to be notice when using matching with perfect observation is we need to be attention to the stratified group where the joint distribution of treatment status and stratification variable equal to zero. This means no one is that group, and the estimation of ATE, ATT and ATC will be biased. Of course, depends on which group is missing, we could still get a consistent estimator of ATT or ATC. For example, if no one in subgroup of treatment group, we could still arrives a ATT, but not ATC.

We could plot the distribution of propensity score for treatment and control group after matching to detect whether their is a unmatched group. If the distribution differs substantially, then it is likely that treatment and control group lack of the common support.

5.1 Propensity score matching and weighting

Exact matching is not practical when X is continuous or contains many variables. Instead, it is possible to first estimate the propensity score, and then compare observations which have a similar propensity score. It is often easier to estimate non-parametrically or semi-parametrically the propensity score than to directly condition on observables. Our critical assumption for matching change to

$E\left[Y^T \mid T=1, P(T=1|X)\right] = E\left[Y^T \mid T=0, P(T=1|X)\right]$

$E\left[Y^C \mid T=1, P(T=1|X)\right] = E\left[Y^C \mid T=0, P(T=1|X)\right]$

where $P(T=1|X)$ is the propensity of being assigned in treatment. Usually this propensity score is estimated by logistic regression.

Treatment and control cases may be missing at random within some of the strata defined by X, such that some strata contain only treatment or only control cases. In this situation, some within-stratum causal effect estimates cannot be calculated with the available data. A set of weighting estimators that rely on estimated propensity scores to solve the sparseness problems that afflict samples of finite size.

The first one is propensity score weighting, which used a set of observed variable to predict the propensity of being treated. But attention: First, this solution only holds in expectation over repeated samples. Second, without a perfect specification of the propensity-score-estimating equation, one cannot rest assured that consistent and unbiased estimates can be obtained. The second point is critical.

5.2 Matching versus OLS

1. They are the same if the treatment effects are constant

2. If treatment effects are different, they will be different, because they apply a different weighting schemes. OLS is effcient under the assumption that the treatment effect is constant, so it weights observation by the conditional variance of the treatment status.

3. Matching does not use cells where there are only treatment observations, whereas OLS takes advantage of the linearity assumption to use all the variables: the treatment group and the control groups may be very dissimilar in matching and in OLS. Matching will throw away all the control observations for which we cannot find at least one treatment observation with the same characteristics.

As I mentioned above, regression specification assume a constant ATE. However, in the presence of the individual level heterogeneity of causal effects, even if the only variable that needs to be adjusted for is given a fully flexible coding, OLS can yield misleading causal effect.

In these cases, least squares estimators implicitly invoke conditional-variance weighting of individual-level causal effects. This weighting scheme generates a conditional-variance-weighted estimate of the average causal effect, which is not an ATE we interested.

Regression estimators with fully flexible coding of the adjustment variables do provide consistent and unbiased estimates of the ATE if either

(1) the true propensity score does not differ by strata or

(2) the average stratum-specific causal effect does not vary by strata.

The first condition would almost never be true (because, if it were, one would not even think to adjust for X because it is already independent of T). And the second condition is probably not true in most applications, because rarely are investigators willing to assert that all consequential heterogeneity of a causal effect has been explicitly modeled. In this case, regression only provide conditional-variance-weighted estimate of the average causal effect.

One thing need to be notice: As in the matching, regression will implicitly drop strata for which the propensity score is either 0 or 1 in the course of forming its weighted average. As a result, a researcher who interprets a regression result as a decent estimate of the ATE, but with supplemental conditional-variance weighting, may be entirely wrong. No meaningful average causal effect may exist in the population. But ATT or ATC is still interpretable under some context.

ATE could get using regresson in a satuarated model where treatment variable interact with each adjustment variable with fully stratified.

Sometimes matching on observables might lead to a greater bias than OLS, if matching is not truly random conditional on observables i.e. matching may not eliminate the omitted variables bias due to unobservables.

6 Difference in Difference

Basic set up, two groups, two time periods. At t = 1, neither group is treated and in period t = 2, one of the groups is treated. We could use the simple first difference to estimate the mean difference between treatment and control groups, but the problem is the estimate is unbiased ATE only under the very strong assumption that, absent the treatment, there would have been no change in average Y.

We could assume the "parallel trend" which means trend in control is equal to what trend would have been among treated had treatment never been applied. We could write the basic set up in the table:

	Treatment	Control
Pre	$E\left(Y_{i1}(0) \mid T_i = 1\right)$	$E\left(Y_{i1}(0) \mid T_i = 0\right)$
Post	$E\left(Y_{i2}(1) \mid T_i = 1\right)$	$E\left(Y_{i2}(0) \mid T_i = 0\right)$

The non-parametric DID estimator under the parallel assumption is

$$DID = \underbrace{(E(Y_{i2}(1)|T_i = 1) - E(Y_{i1}(0)|T_i = 1))}_{\text{Treatment Difference}} - \underbrace{(E(Y_{i2}(0)|T_i = 0) - E(Y_{i1}(0)|T_i = 0))}_{\text{Control treatment}}$$

We also assume constant treatment effects, τ. Then we could define the time effect as λ_t, group effect as $\mu_{i,\{C,T\}}$, where C and T represent control or treatment groups. We could rewrite the table as

	Treatment	Control
Pre	μ_{iT}	μ_{ic}
Post	$\mu_{iT} + \lambda_t + \tau$	$\mu_{iC} + \lambda_t$
Diff	$\lambda_t + \tau$	λ_t
DiD	ATE $= \tau$	

Regression counterpart when idiosyncratic errors are independent of the treatment is

$Y_{it} = \mu_{iC} + \lambda_t t + \gamma T + \tau T \times t + \varepsilon_{it}$

To further understand the above equation, let us decompsite it as

Control group at period 1 is: μ_{iC}

Control group at period 2 is: $\mu_{iC} + \lambda_t$

Treatment group at period 1 is: $\mu_{iC} + \gamma$

Treatment group at period 2 is: $\mu_{iC} + \lambda_t + \gamma + \tau$

DID is: τ

For more than two group and two time periods, we need to add group fixed effect and time fixed effect.

6.1 Checks of DiD strategy

1. Use data for prior periods (say period -1) and redo the DD comparing year 0 and year -1 (assuming there was no policy change between year 0 and year -1). If this placebo DiD is non zero, there are good chances that your estimate comparing year 0 and year 1 is biased as well. More generally, when many years are available, it is very useful to plot the series of average outcomes for Treatment and Control groups and see whether trends are parallel and whether there is a sudden change just after the reform for the Treatment group.

2. Use an alternative control group $C\prime$. If the DiD with the alternative control is different from the DiD with the original control C, then the original DiD is likely to be biased

3. Replace Y by another outcome $Y\prime$ that is not supposed to be affected by the reform. If the DiD using $Y\prime$ is non-zero, then it is likely that the DiD for Y is biased as well.

6.2 Threats to identification and how to deal with it

The treatment needs to be independent of the idiosyncratic shocks so that the variation of the outcome is the same for the treated and control groups, but this might not be plausible. For example in the labor economic, Ashenfelter's dip: which is a empirical finding that people who enroll in job training programs see their earnings decline prior to that training. Thus, the independence of the treatment and idiosyncratic shocks might only hold conditional on covariates.

The non-parametric DID estimator under the parallel assumption and conditional on covariates is:

$$DID = \underbrace{(E(Y_{i2}(1)|T_i = 1, X) - E(Y_{i1}(0)|T_i = 1, X))}_{\text{Treatment Difference}} - \underbrace{(E(Y_{i2}(0)|T_i = 0, X) - E(Y_{i1}(0)|T_i = 0, X))}_{\text{Control treatment}}$$

Regression counterpart when idiosyncratic errors are independent of the treatment is

$Y_{it} = \mu_{iC} + \lambda_t t + \gamma T + \tau T \times t + X_i'\beta + \varepsilon_{it}$

The closer are the Treatment and Control groups, the more convincing is the DD approach. It is often useful to perform simple differences between Treatment and Controls along covariates to see whether Treatment and Controls differ systematically.

DiD is very sensitive to function form such as log transformation of dependent variable. In this case, we could use dynamic linear model as an alternative method.

DiD estimates are more reliable when you compare outcomes just before and just after the policy change because the identifying assumption (parallel trends) is more likely to hold over a short time-window. With a long time window, many other things are likely to happen and confound the policy change effect. However, for policy purposes, it is often more interesting to know the medium or long term effect of a policy change. In any case, one must be very cautious to extrapolate short-term responses to long-term responses.

It very possible that the DiD specification will have serial correlation, therefore, it better to cluster standard errors at the group level. However, cluster standard errors is not correct when the number of clusters is small. In this case, we need to use Moulton (1986) method. Another way is to use block bootstrap.

7 Standard Synthetic Control

The standard synthetic control method is used to identify the causal effect of single treatment observations with many control observations. A classical example is the fall of Berlin wall on economic performance of West Germany.

Basic setup:

1. Suppose 1 treated and J potential control regions
2. Let $W = (w_1, ..., w_J)\prime$ be a vector of non-negative weights for each control region. Let $\sum_k w_j = 1$
3. Control region outcomes are combined using W to create a synthetic counterfactual for the treated region. We want to choose W^* to create the best possible synthetic counterfactual.
4. Let X_1 be covariates for the treated region and X_0 a matrix of covariates for the control region
5. Choose W^* to minimize weighted squared distance,
   $(X_1 - \mathbf{X}_0 W)' \mathbf{V} (X_1 - \mathbf{X}_0 W)$
   where V weights the different covariate discrepancies on the basis of their relative "importance."
6. You are free to choose V as you see fit. For example, Abadie & Gardeazabal (2003) choose V to give priority to minimizing distance between the pre-conflict GDP trend.
7. Then, synthetic control outcomes for the treated region are computed as $\hat{Y}_{it}^C = Y_{jt}' W^*$

This is a booming area, see Yiqing Xu(2017) on Generalized Synthetic Control Method. Yiqing Xu and Xun Pang(2020) on Bayesian Alternative to Synthetic Control, Alberto Abadie(2020) on a nice summary of this method.

8 Fixed Effects Model

The basic idea is that ignorability doesn't hold, conditional on the observed covariates, $Y_{it}(d) \not\perp\!\!\!\perp D_{it} \mid X_{it}$, but ignorability might hold conditional on some unobserved, time-constant, variable: $Y_{it}(d) \not\perp\!\!\!\perp D_{it} \mid X_{it}, \mu_i$. In the fixed effect model, we assume a strict ignorability, which means potential outcomes are independent of the entire history of treatment conditional on the history of covariates and the time-constant heterogeneity.

The two way fixed effect model could be specified as follow:

$$Y_{i,t} = \beta_1 X_{i,t} + \mu_i + \lambda_t + \epsilon_{i,t} \tag{1}$$

where X_i is a vector of independent variables including our core independent variable. μ_i is group time-constant heterogeneity, λ_t is year fixed effect, and ϵ_i is the error term. In some model specifications, we could add time as a variable to control for linear time trend, but this approach is more useful when T is large.

For fixed effect model, we also make similar assumptions as when we are using ordinary least square estimation such as the strictly exogeneity of error term: $E(\varepsilon_{i,t}|X_i, u_i) = 0$[2], no perfect multicollinearity, observations are i.i.d. draws from their joint distribution, idiosyncratic error term is normally distributed. The major difference between OLS and fixed effect model is that in the fixed effect model we allow the group level unobserved time-invariant heterogeneities correlated with independent variables: $cov(u_i|X_i) \neq 0$, which implies that OLS assumption of strictly exogeneity is violated and its estimation is biased.

$$(Y_{i,t} - \bar{Y}_i) = \beta_1(X_{i,t} - \bar{X}_i) + (\mu_i - \bar{\mu}_i) + (\varepsilon_{i,t} - \bar{\varepsilon}_i) \tag{2}$$

Through the demean process in the fixed effect model as show in equation (2), we could eliminate the group level unobserved time-invariant heterogeneities so that strictly exogeneity of error term could be sustained. However, one big disadvantage of this demean process is that we also eliminate time-invariant observable independent variables. Another drawbacks (depends on your research questions) of the fixed effect model is that if you want to analysis the causal path way of your independent variable to dependent variable through a mediation variable, you need all variables to be time-varing variables (or to say level 1 variables), because if the mediation variable is the time-invariant variable (level 2 variable), it will be eliminated during the demean process[3].

If we want to know the effect of time-invariant observable, we could add interaction terms between time-invariant observable and time dummies.

Least Squares Dummy Variable is algebraically equivalent to the within estimator for estimates, but their standard errors will be slightly different. Standard errors from OLS on the within estimator will be slightly off due to incorrect degrees of freedom(but Stata program will automatically adjust this bias, so no worry). Least Squares Dummy Variable get the right standard error but doing this with large clusters can be computationally demanding.

For panel data, it's always better off to use clustered standard errors.

[2]strictly exogeneity assumption could be derived from strict ignorability

[3]McNeish, D., & Kelley, K. (2019). Fixed effects models versus mixed effects models for clustered data: Reviewing the approaches, disentangling the differences, and making recommendations. Psychological Methods, 24(1), 20.

9 Random Effects Model

The random effects model maintain the same model specification as in the fixed effect model. The major difference, however, is now I make additional assumption that $E(u_i|X_i) = E(u_i) = 0$. Random effects model's unobservable heterogeneity assume random draw from population which have a normal distribution, but FE has no such assumption. For the variance and covariance structural, we could further assume $var(u_i) = \sigma_\mu^2$ and $var(\varepsilon_{i,t}) = \sigma_e^2$, so there is no auto-correlation of error terms.

Given the above assumption, the random effect model could be estimated using "partial demean" method:

$$(Y_{i,t} - \theta\bar{Y}_i) = (\beta_0 - \theta\beta_0) + \beta_1(X_{i,t} - \bar{X}_i)' + (\mu_i - \theta\bar{\mu}_i) + (\varepsilon_{i,t} - \theta\bar{\varepsilon}_{i,t}) \tag{3}$$

where $\theta \in [0, 1]$, random effect is more like a weighted average between FE and POLS, and the weight depends on how much variation in error term comes from unit level heterogeniety, if the unit level heterogeniety is large, then RE converge to FE where $\theta = 1$, otherwise, RE converge to POLS where $\theta = 0$. $\theta = 1 - \left(\frac{\sigma_u}{\sqrt{T\sigma_e^2+\sigma_u^2}}\right)$. From the θ equation, we could also see that when T goes large, the fixed effect and random effect converge.

Random effect model is more efficient when data has many clusters, and we need to use clustered standard error.

9.1 FE versus Random effects model

From the above discussion, we know fixed and random effects model impose different identification assumptions. But the core difference is the random effects model assume $E(u_i|X_i) = E(u_i) = 0$. How do we this is true? If true (and all the RE assumptions hold), then RE and FE are consistent, but RE is efficient. If false, then RE is inconsistent, but FE is consistent.

Hausman test uses these facts to develop a hypothesis test of the assumption. If FE and RE estimates are similar then RE assumption is plausible. If not, RE assumption perhaps not valid.

Formally, the Hausman test is

$H = \frac{\widehat{\beta}_{FE} - \widehat{\beta}_{RE}}{\left(\widehat{SE}[\widehat{\beta}_{FE}]^2 - \widehat{SE}[\widehat{\beta}_{RE}]^2\right)^{1/2}}$

where $\widehat{\beta}$ is the estimation from FE and RE accordingly, and $\widehat{SE}[\widehat{\beta}]$ is the standard error from each model, and H has a χ^2 distribution. Under the null hypothesis that RE is correct, H is asymptotically normal. When$\widehat{\beta_{FE}}$ and$\widehat{\beta_{RE}}$ are very different relative to their uncertainty, H will be big in absolute value and we will reject the null. Finally, one thing need to be notice that we must maintain homoskedastic for both models to use Hausman test.

Practically, if T goes to very large, then FE and RE converge. RE unobservable heterogeneity assume

random draw from population which have a normal distribution, but FE has no such assumption. If we are interested in the population distribution of heterogeniety, then random effect will be suitable.

10 Hybrid Model

Although fixed effects model is very useful when there are unobervable time-invariant heterogeneities that correlated with independent variables, it is also limited our ability to say anything about the effects of time-invariant observable variables. Random effects model could add time-invariant independent variables, but its assumptions that $E(\mu_i|X_i) = E(\mu_i) = 0$ is not correct in most of our cases. Also, traditional random effect model as I mentioned above assumes within effects equal to between effect. This assumption may or may not valid. Moreover, random effects model (and mixed effects model) are more efficient when the clusters are above certain threshold (30 usually) [4].

Hybrid model could take the advantages of both fixed effects model and random effects model so that we could get both the within effects and between effect, and estimate the coefficient on time-invariant variables. The basic set up of Hybrid model based on random effect model, but depends on the research question, we could use Mundlak approach or Bell and Jones within between approach. Mathematically, we could eventually get the same results use either methods, the difference only depends on we care about contextual effects or between effects[5].

For Mundlak approach, I specified as follow:

$$Y_{i,t} = \beta_0 + \beta_{1W} X_{i,t} + \beta_{1C} \bar{X}_i + \mu_i + \lambda_t + \epsilon_{i,t} \tag{4}$$

where $\bar{X}_i$ represent the city mean for each independent variables. The rest of the variables stay the same as in fixed effect model.

The within between approach is specified as:

$$Y_{i,t} = \beta_0 + \beta_{1W} X_{i,t,DV} + \beta_{1B} \bar{X}_i + \mu_i + \lambda_t + \epsilon_{i,t} \tag{5}$$

where $X_{i,t,DV}$ is the mean deviate of independent variables, and $\bar{X}_i$ represent the mean of each variables.

Theoretically, we call β_{1C} "contextual effects" while we call β_{1B} between effects. Mathematically, $\beta_{1W} + \beta_{1C} = \beta_{1B}$[6].

[4] McNeish, D., & Kelley, K. (2019). Fixed effects models versus mixed effects models for clustered data: Reviewing the approaches, disentangling the differences, and making recommendations. Psychological Methods, 24(1), 20.

[5] Bell, A., & Jones, K. (2015). Explaining fixed effects: Random effects modeling of time-series cross-sectional and panel data. Political Science Research and Methods, 3(1), 133-153.

[6] Bell, A., Fairbrother, M., & Jones, K. (2019). Fixed and random effects models: making an informed choice. Quality Quantity, 53(2), 1051-1074.

The major advantage in these hybrid model (assume another assumptions for RE model is valid) is once we control for group mean for those time-varying variables, the core assumption for random effect model $E(\mu_i|X_i) = 0$ will be valid. Meanwhile, we need assume that time invariant variables is not correlated with μ_i. I do not have time-invariant variables in my model, so I do not need to concern this problem. But in general, if the correlation between μ_i and time invariant variables is not zero, this will cause problems.

11 Dynamic Model

Dynamic models contain one or more lagged dependent variables, allowing for the modeling of a partial adjustment mechanism. In a dynamic panel data model, we consider that each unit in the panel is following a trajectory, adjusting its outcome variable toward a target for example a desired capital stock for a firm, or a desired stock of retirement assets for a household. As long as we observe the panel units more frequently than they can fully adjust to their target value, we are observing disequilibrium values, and the prior value(s) of the outcome variable are one of the factors driving that trajectory.

The model specified as follow:

$$Y_{i,t} = \beta_0 + \beta_1 Y_{i,t-1}) + \beta_2 X_{i,t} + \mu_i + \lambda_t + \epsilon_{i,t} \tag{6}$$

Because your covers a short time period, a small T, you can not use fixed effects estimator as it will introduce endogeneity problem (Nickell bias). When we demean the error term as in equation (2), the error term $(\varepsilon_{i,t} - \bar{\varepsilon}_i)$ will be correlated with $(Y_{i,t-1} - \bar{Y}_{i,t-1})$. However, theoretically, when T goes very large, the mean of error term will goes to zero if there is no auto-correlation, and the fixed effect estimator will be appropriate.

To estimate the the above dynamic model, I will use the Arellano-Bond two step "Generalized Method of Moments" estimator. Basic, we need to first take the first difference for dependent and independent variables and use the further lags of Y as instruments for the lagged difference term.

For Arellano-Bond two step estimator, I rely on the assumption that there is no second order auto-correlation between my error terms and the instrument set being truly exogenous. I will empirical test these assumption when I discuss my model results.

12 Multi-Level Model

To demonstrate that within between hybrid model is conceptually the same as multi-level model, I will specify the multi-level model as follow:

$$\text{First Level}: Y_{i,t}) = \beta_{0,i} + \beta_{1W} X_{i,t,DV} + \lambda_t + \epsilon_{i,t} \tag{7}$$

$$\text{Second Level}: \beta_{0,i} = \beta_{0,0} + \beta_{1B} \bar{X}_i + v_{0i} \tag{8}$$

where all the variables maintain the same definition as before, and $v_{0,i}$ is the error term of second level model. For both the first and second level, we assume strict exogenous of the error terms, and the error term are normally distributed. Here $v_{0,i}$ is a random effect for the ith city that captures how much the intercept for the ith city differs from the overall intercept $\beta_{0,0}$.

Multi-level modeling have the same advantages as in the REWB model, and we could easily model data have multiple levels of clustering such as firms nested within cities nested within province.

If we substitute equation (8) to (7), then we get exactly same equation as (5). This is mathematically showing that multi-level model is conceptually the same as hybrid model.

13 Instrumental Variable

The OLS estimate of β is biased when the exgeniety assumption fails. This may caused by omitted variables, measurement error, or reverse causality. A way to solve this problem to use instrumental variables approach. Instrument variable is very hard to find and convince other readers, but a good instrument need to meet the following requirement:

1. Exclusion restriction: no common cause of instrument variable and outcome variable, and no direct or indirect effect of the instrument on the outcome not through the treatment.

2. Relevant: in the first stage, instrument or instruments need to significant or jointly significant.

If the above requirement satisfied, then the ATE when we have instrument Z and assume constant treatment effect is:

$Y_i = \alpha + \delta T + \varepsilon$, and $E(\varepsilon|T) \neq 0$, then

$\text{Cov}\,[Y_i, Z_i] = \text{Cov}\,[\alpha + \delta T + \varepsilon, Z_i] = \delta\,\text{Cov}\,[T_i, Z_i]$

$\delta = \frac{\text{Cov}[Y_i,Z_i]}{\text{Cov}[T_i,Z_i]} = \frac{\frac{\text{Cov}[Y_i,Z_i]}{\text{Var}[Z_i]}}{\frac{\text{Cov}[T_i,Z_i]}{\text{Var}[Z_i]}} = \frac{\text{Reduced form}}{\text{First stage}}$

when the instrument is binary, we could derive the wald estimator:

$\delta = \frac{\text{Cov}(Y_i,Z_i)}{\text{Cov}(T_i,Z_i)} = \frac{E[Y_i|Z_i=1]-E[Y_i|Z_i=0]}{E[T_i|Z_i=1]-E[T_i|Z_i=0]}$

Weak instrument

For identification, we need to the fist stage to be strong, this means for fisrt stage

$T = Z\gamma + v$

γ need to be none zero and significant, or "first stage F statistical larger than 10" on the instrumental variables (Stock Watson, 2007, based on bias bounds), otherwise, the instrument is weak. When we have weak instrument:Cov $(T_i, Z_i) \approx 0$, the estimated ATE will be very volatile. Moreover, when the instrument is "imperfect", Cov $(\varepsilon_i, Z_i) \neq 0$, then

$\text{Cov}\left[Y_i, Z_i\right] = \text{Cov}\left[\alpha + \delta T + \varepsilon, Z_i\right] = \delta\,\text{Cov}\left[T_i, Z_i\right] + \text{Cov}\left(\varepsilon, Z_i\right)$

$\frac{\text{Cov}(Y_i,Z_i)}{\text{Cov}(T_i,Z_i)} = \delta + \frac{\text{Cov}(\varepsilon,Z_i)}{\text{Cov}(T_i,Z_i)}$

the above equation means when we have weak instrument, then even very small violations of the exclusion restriction can lead to large inconsistencies. Meanwhile, in the small sample, the IV estimator biased toward OLS when we have weak instrument. Also, in the small sample, if we add too many instrument variables at the first stage, then fist stage R^2 will be approach to 1 when instrument variable sets is very large, then at this case, IV converge to OLS. But we are rarely confront this situation.

But notice, weak instrument is not the end of the world. There are some ways to better handle weak instruments see Andrews, Stock, and Sun(2019) paper.

13.1 2SLS

When we have other exogenous control variables, we could use 2SLS estimation for instrumental variable approach. Generally, it is

1. Run regression of treatment on covariates and instrument
2. Construct fitted values of treatment
3. Run regression of outcome on covariates and fitted values

Notice, it's better to use ivreg in Stata instead of doing it by hand because the standard error is not correct from the above procedure.

Mathematically, we could derive the 2SLS estimator as follow:

1. From the first stage, $X = Z\gamma + v$, where X is the endogenous variable, Z include covariates and instruments, we could get the $\hat{\gamma} = (Z'Z)^{-1} Z'X$, and the fitted value as $\hat{X} = Z\hat{\gamma}$
2. take the fitted values, $Z\hat{\gamma}$ and multiply both sides of the outcome equation $Y = X'\beta + \varepsilon$ by them:

$Z\hat{\gamma}Y = Z\hat{\gamma}X'\beta + Z\hat{\gamma}\varepsilon$

$E(Z\hat{\gamma}Y) = E(Z\hat{\gamma}X'\beta) + E(Z\hat{\gamma}\varepsilon)$

3. it is easy to show that the last term of the above equation equal to 0, therefore,

$$E\left[Z\hat{\gamma}Y\right] = E\left[Z\hat{\gamma}'X'\right]\beta$$
$$E\left[\hat{X}Y\right] = E\left[\hat{X}X'\right]\beta$$
$$\beta = (E\left[\hat{X}X'\right])^{-1}E\left[\hat{X}Y\right]$$
$$\hat{\beta} = (\hat{X}X')^{-1}\hat{X}Y \text{ where } \hat{X} = Z\left(Z'Z\right)^{-1}Z'X$$

Use the same procedural as in the OLS, we could also derive the variance of $\hat{\beta}$, which is the following:
$\widehat{\text{var}}(\hat{\beta}) = \hat{\sigma}^2\left(\hat{\mathbf{X}}'\hat{\mathbf{X}}\right)^{-1}$
where $\hat{\sigma}^2 \equiv (N-K)^{-1}\sum_{i=1}^{N}\hat{\varepsilon}_i^2$
and the robust standard error as:
$\widehat{\text{var}}(\hat{\beta}) = \left(\widehat{\mathbf{X}}'\widehat{\mathbf{X}}\right)^{-1}\left(\sum_i \hat{\varepsilon}_i^2\widehat{X}_i\widehat{X}_i'\right)\left(\widehat{\mathbf{X}}'\widehat{\mathbf{X}}\right)^{-1}$
where $\hat{\varepsilon}_i = Y_i - X_i'\hat{\beta}$

Overidentification

Instrumental variables approach rely on the critical assumption of exclusion restriction, however, this cannot be tested and has to be assessed on a case by case basis. When we have more instruments than endogenous variables than we are overidentified. With more than one instrument and constant effects, we can test for the plausibility of the exclusion restriction(s) using an overidentification test.

Sargan-Hausman test:

1. Under the null of all valid instruments, using all instruments versus a subset should only differ by sampling variation.

2. Regress 2SLS residuals, $\hat{\varepsilon}_i$ on X_i and calculate R^2 from this regression

3. Under the null and homoskedasticity, NR^2 has χ^2_m where the degree of freedom m depends on how many overidentifying restrictions there are

If we reject the null hypothesis in these overidentification tests, then it means that the exclusion restrcitions for our instruments are probably incorrect. Note that it won't tell us which of them are incorrect, just that at least one is.

These tests are useful, but have two problems. These overidentification tests depend heavily on the constant effects assumption. Their power is not very strong and they tend to accept too often

13.2 Instrumental variables with heterogenous effects

Under the potential outcome frame, the instrument variable approach when we have heterogenous effects need to have the following assumption:

1. Independent: $\{\mathrm{Y}_i\left(\mathrm{T}_{1i}, 1\right), \mathrm{Y}_{0i}\left(\mathrm{T}_{0i}, 0\right), \mathrm{T}_{1i}, \mathrm{T}_{0i}\} \perp\!\!\!\perp \mathrm{Z}_i$

   Which means the instrument is independent of potential outcomes and potential treatment assignment. We can weaken this to conditional ignorability. Based on this assumption, we could estimate the Intent to Treat(ITT): $E\left[Y_i \mid Z_i = 1\right] - E\left[Y_i \mid Z_i = 0\right] = E\left[Y_i\left(T_i(1), 1\right) - Y_i\left(T_i(0), 0\right)\right]$

2. Exclusion Restriction: The instrument has no direct effect on the outcome, once we fix the value of the treatment. Given this exclusion restriction, we know that the potential outcomes for each treatment status only depend on the treatment, not the instrument: $Y_i(d, 1) = Y_i(d, 0) \quad \text{for } d = 0, 1$

3. First stage. the instrument must have an effect on the treatment

4. Monotonicity: this is introduced by Imben and Angrist(1994) means that while instrument may have no effect on some people, all of those who are effected are effected in the same way, although the effects could vary across the population $\mathrm{T}_{1i} \geq \mathrm{T}_{0i}$ or $\mathrm{T}_{1i} \leq \mathrm{T}_{0i}$ for all i.

   This assumption sometimes called no defiers assumption. The monotonicity assumption remove the possibility of there being defiers in the population. Anyone with $T_i = 1$ when $Z_i = 0$ must be an always-taker and anyone with $T_i = 0$ when $Z_i = 1$ must be a never-taker.

Under these four assumptions, the Wald estimator is equal what we call Local average treatment effect (LATE) or the complier average treatment effect (CATE). This is the ATE among the compliers: those that take the treatment when encouraged to do so. LATE is

$\frac{E[Y_i|Z_i=1]-E[Y_i|Z_i=0]}{E[T_i|Z_i=1]-E[T_i|Z_i=0]} = E\left[Y_i(1) - Y_i(0) \mid T_i(1) > T_i(0)\right]$

This is a extremely important result. It shows once we allow for heterogeneous effects, all we can estimate with IV is the effect of treatment among compliers. Without further assunmption, LATE$\neq$ ATE. Complier group depends on the instrument, different IVs will lead to different estimands. 2SLS "cheats" by assuming that the effect is constant, so it is the same for compliers and non-compliers.

Notice, with one-sided noncompliance, the ATT is the same as the LATE.

13.3 Identification checks and Falsification tests

Given the LATE assumption, we could use Kolgomorov-Smirnov statistic or Huber Mellace (2015) to test whether these assumptions are valid.

The exclusion restriction cannot be tested directly, but it can be falsified. Falsification test Test the reduced form effect of Z_i on Y_i in situations where it is impossible or extremely unlikely that Z_i could affect T_i. Because Z_i can't affect T_i, then the exclusion restriction implies that this falsification test should have 0 effect. Nunn and Wantchekon (2011): use distance to coast as an instrument for Africans, use distance to the coast in an Asian sample as falsification test.

13.4 Size, characteristics of the compliers

While we cannot identify who is a complier and who is not a complier in general, we can estimate the size of the complier group:

$\Pr\left[T_i(1) > T_i(0)\right] = E\left[T_i(1) - T_i(0)\right] = E\left[T_i \mid Z_i = 1\right] - E\left[T_i \mid Z_i = 0\right]$

Can extend this to calculate features of the complier group: covariate means, variances, etc. Abadie (2003) shows how to weight the data to estimate these quantities.

13.5 Multiple instruments under LATE

Because under heterogenous effects, different instrument estimate different LATE. 2SLS estimator using these different instruments is a weighted sum of the two component LATEs. For the first stage when we have two instrument:

$\widehat{T}_i = \pi_1 Z_{1i} + \pi_2 Z_{2i}$

$\delta_{2SLS} = \psi\delta_1 + (1-\psi)\delta_2$

$\psi = \frac{\pi_1 \operatorname{Cov}(T_i, Z_{1i})}{\pi_1 \operatorname{Cov}(T_i, Z_{1i}) + \pi_2 \operatorname{Cov}(T_i, Z_{2i})}$

Thus, the 2SLS estimate is a weighted average of causal effects for each instrument, where the weights are related to the strength of first-stage.

When the above independent assumption only hold conditional on covariates, conditional independent. Under these assumptions, with fully saturated first and second stages, then 2SLS estimates a weighted average of the covariates-specific LATEs (very similar to regression). Abadie (2003) shows how to estimate the overall LATE using a weighting approach based on a "propensity score" for the instrument.

14 Regression Discontinuity Design

RDD can be used when the treatment is a discontinuous function of an underlying continuous variable. When this rule is followed at least approximately, it means that two subjects with very close characteristics will be exposed to different treatments. The idea of RDD is to compare outcome for people whose value of the underlying targeting variable is just below and just above the discontinuity.

In RDD, subjects being assigned to the treatment condition is not the same as receiving or complying with the treatment. Any RD design where compliance with treatment assignment is imperfect is referred to as Fuzzy RD design, and sharp RD design is where treatment condition assigned is identical to the treatment condition actually received for all units.

14.1 Sharp RDD

Formally, the sharp RD effect can be defined as follow:

Imagine first that treatment rule is based on a running variable X, which means treatment is then deterministic and discontinuous in X and that the treatment rule is:

T=1 if X $\geq \bar{X}$

T=0 if X $< \bar{X}$

Then with a large sample, you would compute (for some ϵ):

$E[Y \mid \bar{X} \leq X < \bar{X}+\epsilon]-E[Y \mid \bar{X}-\epsilon \leq X < \bar{X} = E\left[Y^T \mid T, \bar{X} \leq X < \bar{X}+\epsilon\right]-E\left[Y^C \mid C, \bar{X}-\epsilon \leq X < \bar{X}\right]$

The assumption is that as the ϵ goes to 0, the difference between the two groups in the absence of the treatment shrinks to 0. One critical assumption of RDD is function need to be continuous around the cutpoint.

Notice, the treatment rule implies no overlap in treated and control observations over X. This is totally different from conditional independent assumption where we assume positivity.

The results captures a causal effect that is local in nature. And from some perspective, this parameter can not even be interpreted as causal because it can not be reproduced via manipulation or experiment. Therefore. the RD treatment effect tends to have limited external validity, that is the RD effect is generally not representative of the treatment effects that would occur for units with score away from the cutoff. RD design is only the average effect of treatment for units local to the cutoff.

If the potential outcomes change at the discontinuity for reasons other than the treatment, then smoothness will be violated. For instance, if people sort around threshold, then you might get jumps other than the one you care about. If things other than the treatment change at the threshold, then that might cause discontinuities in the potential outcomes.

14.2 RDD plot

For visualization, RD plot present two summaries: 1. a global polynomial fit and local sample means.

Three are two general ways to plot the scatter plot, one is even space bin plot and the other one is equal quantile. Thr former one have the same length and the later one contain the same number of observations.

In practice, you could select your preferred bin number, but there are ways to define the optimal bin number under equally spaced and equally quantile method.

From the RD plot, we could see if obvious discontinuity at the threshold and obvious unexplained discontinuities.The formal statistical analyses discussed are essentially just sophisticated versions of RD plot, and if the basic plot does not show any evidence of a discontinuity, there is relatively little chance that the more sophisticated analyses will lead to robust and credible estimates with statistically and substantially significant magnitudes.

The next step is a to plot covariates by the running variable to see if these covariates also jump at the discontinuity. If there are jumps, then it suggests smoothness violated.

We could also plot the density of running variable around threshold point, if there is a jump, then it's possible there is sorting.

14.3 Local Polynomial Point Estimation

1. Choose a polynomial order p and a kernel function $K(\cdot)$

2. Choose a bandwidth h

3. For observations above the cutoff (i.e., observations with $X_i \geq c$), fit a weighted least-squares regression of the outcome Y_i on a constant and $(X_i - c), (X_i - c)^2, \ldots, (X_i - c)^p$, where p is the chosen polynomial order, with weight $K\left(\frac{x_i - c}{h}\right)$ for each observation. The estimated intercept from this local weighted regression, $\hat{\mu}_+$, is an estimate of the point $\mu_+ = E\left[Y_i(1) \mid X_i = c\right]$:

   $\hat{\mu}_+ : \hat{Y}_i = \hat{\mu}_+ + \hat{\mu}_{+,1}(X_i - c) + \hat{\mu}_{+,2}(X_i - c)^2 + \cdots + \hat{\mu}_{+,p}(X_i - c)^p$

4. For observations below the cutoff (i.e., observations with $X_i < c$), fit a weighted least-squares regression of the outcome Y_i on a constant and $(X_i - c), (X_i - c)^2, \ldots, (X_i - c)^p$, where p is the chosen polynomial order, with weight $K\left(\frac{X_i - c}{h}\right)$ for each observation. The estimated intercept from this local weighted regression, $\hat{\mu}_-$, is an estimate of the point $\mu_- = E\left[Y_i(0) \mid X_i = c\right]$

   $\hat{\mu}_- : \quad \hat{Y}_i = \hat{\mu}_- + \hat{\mu}_{-,1}(X_i - c) + \hat{\mu}_{-,2}(X_i - c)^2 + \cdots + \hat{\mu}_{-,p}(X_i - c)^p$

5. Calculate the Sharp RD point estimate: $\hat{\tau}_{\mathrm{SRD}} = \hat{\mu}_+ - \hat{\mu}_-$

The first step that choose kernel function and local polynomial is very important. Most of the time, kernel function may not have a huge impact on estimation. But how you choose local polynomial directly affect the estimation. The choice of bandwidth around the cutoff point is fundamental for the analysis and interpretation of RD designs, as it directly affects the properties of local polynomial estimation and inference procedures, and empirical findings are often sensitive to its particular value.

Smaller bandwidth makes the linear approximation results on a smaller misspecification error. This illustrates the general principle that, given a polynomial order, the accuracy of the approximation can always be improved by reducing the bandwidth. Meanwhile, choosing a smaller bandwidth will tend to increase the variance of the estimated coefficients because fewer observations will be available for estimation. For this reason, the choice of bandwidth is said to involve a "bias-variance trade-off." There are several ways to choose the optimal bandwidth, and we could also construct confidence interval after estimation. A common way to choose the optimal bandwidth is to use the cross validate to determine which choice of bandwidth arrive the minimum mean square error of our estimation function.

Addding covariates

It's possible to add covariates into the basic RD design, and the following specification looks like DID specification. But to do this, the covariate must be balanced around the cutoff point. No treatment effect on covariates which is also to say covariates are pre-determined.

It follows that, in general, it is not possible to include imbalanced covariates in the estimation to "fix" an RD design in which predetermined covariates are discontinuous at the cutoff and the required continuity assumptions are called into question. For the inclusion of covariates to "control for" unexpected imbalances, researchers will either need to invoke parametric assumptions on the regression functions to enable extrapolation, or redefine the parameter of interest. Therefore, analogously to the case of randomized experiments, the generally valid justification for including covariates in RD analysis is the potential for efficiency gains, not the promise to fix implausible identification assumptions. In many RD applications, a covariate-adjusted local polynomial estimation strategy will lead to shorter confidence intervals for the RD treatment effect, increasing the precision of statistical inferences.

14.4 Fuzzy RDD

In the fuzzy RDD, the treatment assignment is no longer a deterministic function of the forcing variable, but there is still a discontinuity in the probability of treatment at the threshold. Fuzzy RD is often useful when the a threshold encourages participation in program, but does not actually force units to participate.

Basically, in the fuzzy RDD, the running variable is an instrument. This is to say the running variable around threshold need to satisfy the instrument variable assumption as I mentioned in section 13.2. And of course, the estimation from fuzzy RDD is LATE, the effect on those compliers.

15 Statistical Machine Learning

High dimensional curse the higher the dimension you have in the data the less data you will use to train your model and the variance and bias of your prediction is higher. This is also related to sparsity. Usually, non-parametric method in classification perform worse than linear model when there are many features in the data.

Re-sampling methods: 1) validation set approach: the easiest way is just to divide the data to two part, one uses for train the model and the rest of it use for test. But this may has high test error rate than the truth. 2) leave one out cross validation: hold one set of observation as test data and estimate the model using the n-1 data, then repeat this procedure and compute $\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\mathrm{MSE}_i$. This method is computatational costly. There is some shortcut when use least sqaure linear and poliynomial, but this shortcut is not generaly true. 3) k-fold validation which divide data to 5 or fold and hold one fold as test and estimate model using k-1 fold, this saves computational power but may lead to greater bias compare to loocv

Non-linear models 1) polynomial regression, 2) step functions, where first divide data to several range and then fit different constant in each bin. The bad side is it is hard to capture the trend if the breaked range is not right. 3) basis function: a more popular method which include the following, 1) piecewise polynomial regression, this is similar to step functions but this time we fit polynomial regression in each interval. More knots leads to more flexible polynomial function. 2) spline, such as cubic and nature spline. We need to gurantee that they are continues at the knots. 3) smoothing spline where we take a penalty for the unsmoothness part of the function with x_i is each knots. 4) local regression